

Enhanced performance of inverted CsPbI₃ perovskite solar cells by using carbohydrazide interface modification

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Chenyu Zhou,¹ Xiaohui Liu,^{1,a)} Jiaying Liu,¹ Yiyang Qiu,¹ Heng Qiao,¹ Hubin Zeng,¹ Jing Zhang,¹ Like Huang,¹ Ning Chen,^{1,a)} and Yuejin Zhu²

AFFILIATIONS

¹School of Physical Science and Technology, Ningbo University, Ningbo 315211, China

²School of Information Engineering, College of Science and Technology, Ningbo University, Ningbo 315300, China

^{a)}Authors to whom correspondence should be addressed: liuxiaohui@nbu.edu.cn and chenning1@nbu.edu.cn

ABSTRACT

Inorganic perovskite CsPbI₃, with its suitable bandgap and excellent thermal stability, is a promising light-absorbing material for perovskite solar cells (PSCs). However, the detrimental defects in CsPbI₃ not only act as recombination centers but also accelerate film degradation, resulting in poor power conversion efficiency (PCE) and reduced stability. Herein, carbohydrazide (CBZ) was employed as an upper surface passivator for CsPbI₃ films. The amino and carbonyl functional groups of CBZ can strongly anchor to undercoordinated Pb²⁺ via tight coordination bonds, effectively passivating surface and grain boundary defects of CsPbI₃. Additionally, CBZ modification can ameliorate film morphology, optimize interfacial energy level alignment, enhance charge extraction, and suppress non-radiative recombination. Consequently, CBZ-modified inverted PSCs yield a champion PCE of 19.76% with an outstanding fill factor (FF) of 85.26%, which is one of the highest FF reported for CsPbI₃ PSCs so far. Moreover, CBZ-modified devices exhibit significantly enhanced long-term storage stability in air. This work highlights the use of molecularly engineered modifiers as a promising approach to boost the efficiency and stability of inverted PSCs, providing a valuable interface design reference.

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Metal halide perovskite solar cells (PSCs) have emerged as one of the most promising next-generation photovoltaic technologies, with the highest power conversion efficiency (PCE) surpassing 27%.^{1,2} Compared to organic-inorganic hybrid perovskites, all-inorganic perovskites without organic cations have garnered significant attention in recent years due to their excellent thermal stability and photostability.^{3–5} Particularly, CsPbI₃ possesses a suitable bandgap and single-halide composition, making it a highly ideal candidate for both single-junction and tandem devices.^{6,7} Substantial progress in fabrication techniques has propelled the PCE of CsPbI₃-based PSCs, reaching 22% over the past few years.^{8,9} Nevertheless, this performance remains notably inferior to the Shockley-Queisser (SQ) limit and state-of-the-art organic-inorganic hybrid counterparts.¹⁰ Moreover, due to its inherently small Goldschmidt tolerance factor and ionic lattice vulnerability, black-phase CsPbI₃ is prone to structural decomposition, and its stability still requires further improvement.¹¹

Despite the development of diverse alternative approaches, solution processing remains the dominant manufacturing route for CsPbI₃

films because of its cost-effectiveness and scalability.^{12–14} However, the rapid crystallization kinetics coupled with high-temperature annealing inevitably introduce various types of defects on the surface and grain boundaries (GBs) of perovskite films, particularly undercoordinated Pb²⁺ and iodine vacancies (V_I) defects caused by insufficient coordination.¹⁵ These undesirable defects usually serve as non-radiative recombination centers, trapping charge carriers and diminishing critical photovoltaic parameters.^{16,17} Concurrently, high concentration of defects can provide penetration pathways for moisture and oxygen molecules, leading to accelerated decomposition of the perovskite structure.¹⁸ To address the above issues, extensive studies have been conducted. For instance, the crystal growth of CsPbI₃ perovskite has been modulated via an additive strategy to produce high-quality thin films with reduced defect densities. In this regard, organic molecules such as hydrazides, silanes, ionic liquids, molten salt, and zwitterions have been incorporated into perovskite precursors, resulting in significant improvements in device performance.^{19–23} However, detrimental defects in solution-processed perovskite films are still difficult to

completely eliminate.²⁴ Therefore, passivating harmful defects via post-treatment on the as-prepared perovskite film surface becomes critically important. Many surface modifiers, including histamine, propylamine hydrochloride, cinnamic acids, and quantum dots, have been efficiently utilized to suppress defects and ameliorate phase integrity.^{25–28} Recently, our group and Fang *et al.* demonstrated that diamine-induced surface polishing or reconstruction mechanisms can remarkably suppress defects and inhibit moisture invasion, greatly promoting device PCE and stability.^{29–31}

Critically, both conventional (n-i-p) and inverted (p-i-n) structured CsPbI₃ devices consistently exhibit limited fill factors (FF), rarely exceeding 85%, which significantly lags behind theoretical value.^{32,33} The FF parameter of solar cells serves as a critical indicator of interfacial charge transport efficiency and non-radiative recombination losses, closely correlating with defect states at perovskite surfaces and interfaces.^{34,35} Thus, efficient carrier transport and reduced non-radiative recombination losses are crucial for achieving high FF values. It is imperative to explore effective defect-passivation strategies with mechanistic understanding, thereby synergistically boosting critical device parameters (particularly FF) and long-term stability.

Herein, we introduced carbohydrazide (CBZ) molecule bearing amino (-NH₂) and carbonyl (C=O) functional groups as an upper surface passivation agent for CsPbI₃ films. The multifunctional groups in CBZ molecule exhibit superior anchoring capacity and can form tight coordination bonds with undercoordinated Pb²⁺ to passivate defects on the CsPbI₃ perovskite surface and GBs. Simultaneously, CBZ modification effectively improves film morphology, optimizes energy level alignment at interfaces, enhances charge transport efficiency, and suppresses non-radiative recombination. Ultimately, the inverted CsPbI₃ PSC with CBZ treatment achieved a champion PCE of 19.76% with an impressive FF of 85.26%. Furthermore, the CBZ-passivated devices exhibited significantly enhanced long-term stability in ambient air.

Figure 1(a) illustrates the chemical structure and electrostatic potential distribution map of a CBZ molecule. Structural analysis reveals two distinctive types of functional groups: two terminal hydrazine (-NH-NH₂, featuring the -NH₂ tail with lone-pair electrons) groups at both ends, and a central carbonyl (-C=O) group in the middle. We speculate that the terminal -NH₂ group from the hydrazine unit, owing to its high electron density and minimal steric hindrance, can preferentially coordinate with undercoordinated Pb²⁺ through

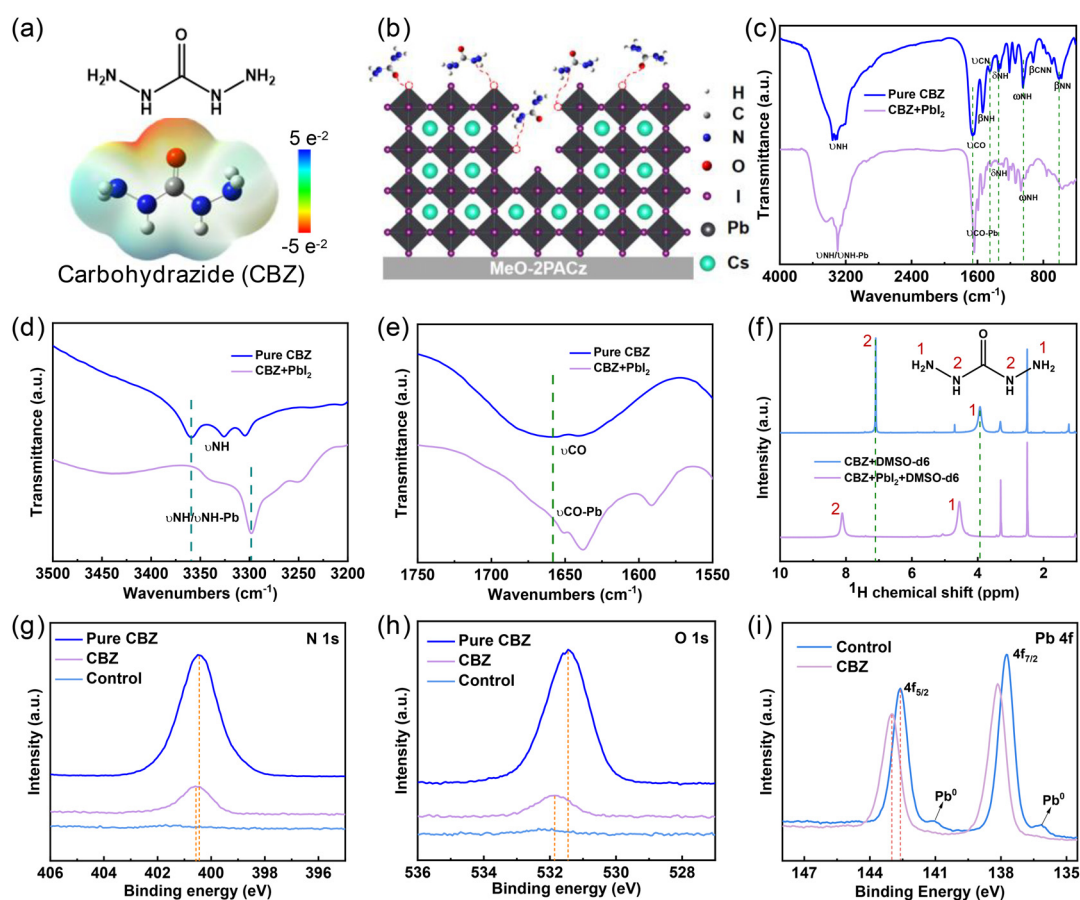


FIG. 1. (a) Molecular structure and electrostatic potential (ESP) of CBZ. (b) Schematic illustration of the passivation mechanism for CBZ on CsPbI₃ surface. (c) FTIR spectra and (d) and (e) zoomed-in regions of pure CBZ, and CBZ+PbI₂ samples. (f) ¹H NMR spectra. XPS spectra of (g) N 1s, (h) O 1s, (i) Pb 4f for control, CBZ-modified CsPbI₃, and pure CBZ films.

Lewis acid–base interactions, serving as the cornerstone of defect passivation;^{36,37} whereas many previous works suggest that the $-C=O$ group can also concurrently passivate lead defects via coordination interactions with Pb^{2+} , not only compensating for sub-nanoscale defects inaccessible to $-NH_2$ but also stabilizing the coordination structure through polar effects to mitigate ligand dissociation.^{38–41} Thus, the synergistic effects among these functional groups maximally cover the uncoordinated lead defects on the $CsPbI_3$ surface, while simultaneously reinforcing perovskite stability through multidentate and cooperative interactions. Therefore, immediately after the preparation of $CsPbI_3$ film, we performed direct post-treatment on the $CsPbI_3$ surface using CBZ modifier for defect passivation. To evaluate whether the CBZ layer undergoes any decomposition at the $CsPbI_3$ interface, we conducted a light-soaking test on bare perovskite films and thermogravimetric analysis (TGA) of pure CBZ molecules. Both the control and CBZ-modified $CsPbI_3$ films exhibit no discernible macroscopic changes in film appearance, suggesting that the CBZ-modified interface does not introduce a conspicuous light-sensitive degradation pathway (Fig. S1). The TGA data show that CBZ has outstanding thermal stability below $160^\circ C$ and is not expected to undergo thermal decomposition under our processing conditions (Fig. S2). The defect passivation mechanism model of CBZ molecule on $CsPbI_3$ perovskite surfaces is schematically illustrated in Fig. 1(b).

To verify our proposed passivation mechanism, we conducted a series of characterizations including Fourier transform infrared spectroscopy (FTIR), liquid-state 1H nuclear magnetic resonance spectroscopy (1H NMR), and x-ray photoelectron spectroscopy (XPS). As the FTIR results presented in Figs. 1(c) and 1(e) and S3 show, the NH stretching vibration (ν_{NH}), bending vibration (β_{NH}), and CO stretching vibration (ν_{CO}) characteristic peaks can be observed for pure CBZ molecule at ≈ 3400 – 3150 , 1536 , and 1652 cm^{-1} , respectively.⁴² Upon

mixing with PbI_2 , the CBZ+ PbI_2 sample exhibits significant ν_{NH} , and ν_{CO} peak shifts, indicating simultaneous interaction of the $-NH_2$ and $-C=O$ moieties of CBZ with undercoordinated Pb^{2+} sites. 1H NMR was further performed to investigate these interactions by dissolving the compounds in deuterated DMSO. The pure CBZ molecule exhibits two distinct hydrogen environments, detected at 3.94 and 7.10 ppm for $-NH_2$ and $-NH-$ groups, respectively [Fig. 1(f)]. After the addition of PbI_2 , the proton signals exhibit notable downfield shifts to 4.55 and 8.11 ppm, respectively, confirming strong interactions between the amino moiety and perovskite components. Moreover, we directly spread CBZ powder onto the prepared $CsPbI_3$ surface (Fig. S4), and found that the perovskite film was subsequently damaged, exhibiting behavior similar to previously reported cases where liquid diamine molecules degraded perovskite films,^{29,31} which further confirms the reactive interaction between CBZ and $CsPbI_3$ perovskite. Additionally, XPS measurements further substantiate the chemical interactions between CBZ molecules and the $CsPbI_3$ film. The characteristic N 1s (400.5 eV) and O 1s (531.4 eV) peaks are observed only in the CBZ-modified film but absent in the control film, suggesting the retention of CBZ on the perovskite surface after modification [Figs. 1(g) and 1(h)]. Notably, these peaks feature a considerable shift compared to their positions in the pure CBZ film, indicating altered chemical environments. The control film initially displays two dominant Pb 4f peaks at 142.6 and 137.8 eV , which shift to higher binding energies upon CBZ modification [Fig. 1(i)], further evidencing the effective coordination between the functional groups of CBZ and undercoordinated Pb^{2+} .^{40,43} Importantly, two minor peaks corresponding to Pb^0 species were detected in the control film, but completely disappeared in the CBZ-modified film. The XPS survey scan and I 3d spectra are provided in Fig. S5. Overall, these results obtained from FTIR, 1H NMR, and XPS measurements demonstrate that the $-NH_2$ and $-C=O$ functional

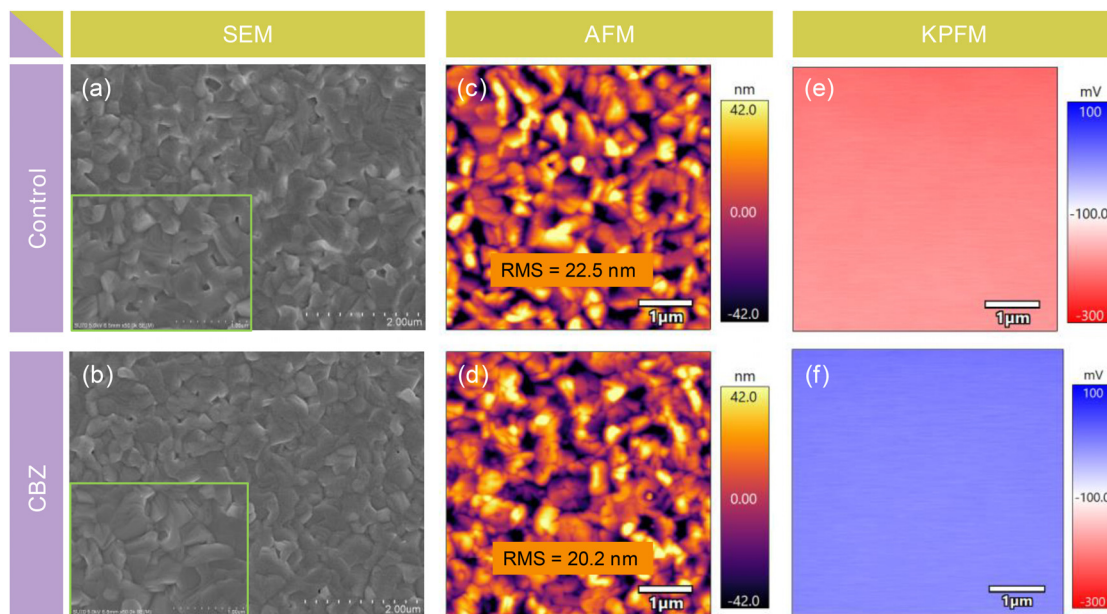


FIG. 2. Top-view SEM images of (a) control and (b) CBZ-modified $CsPbI_3$ films. AFM images of (c) control and (d) CBZ-modified $CsPbI_3$ films. KPFM images of (e) control and (f) CBZ-modified $CsPbI_3$ films.

groups of CBZ can interact with uncoordinated Pb^{2+} species in the CsPbI_3 film, thereby passivating defects and suppressing non-radiative recombination.

Scanning electron microscopy (SEM) was utilized to examine the topographic features of various CsPbI_3 films. As shown in Fig. 2(a), the control CsPbI_3 film exhibits a nonuniform surface morphology with discernible pinhole defects. In contrast, the CBZ-modified film with an optimal concentration of 0.2 mg/mL displays significantly improved crystalline quality, featuring a dense polycrystalline microstructure with negligible pinholes [Figs. 2(b) and S6]. Furthermore, atomic force microscopy (AFM) was employed to analyze the surface profiles of various CsPbI_3 films. The control film presents a large surface roughness with an average root mean square (RMS) of 22.5 nm, whereas the CBZ-modified film features a smoother surface with reduced RMS of 20.2 nm [Figs. 2(c) and 2(d)]. The morphological improvement is advantageous for enhancing interfacial contact between the perovskite and electron transport layer (ETL). Additionally, Kelvin probe force microscopy (KPFM) was utilized to assess the spatial distribution of surface potentials of corresponding films [Figs. 2(e) and 2(f)]. Relative to the control film, the CBZ-modified film exhibits an elevated average surface potential, suggesting a decreased work function and enhanced n-type characteristics. This result can be attributed to the remarkably weakened pinning effect, caused by the coordination of CBZ's functional groups with under-coordinated sites, which effectively passivates the defects.

The effect of CBZ modification on the crystallization of CsPbI_3 thin films was systematically investigated through x-ray diffraction (XRD) measurements. As shown in Figs. 3(a) and S7, all modified films exhibit characteristic diffraction peaks at 14.3° and 28.9° , corresponding to the (110) and (220) crystallographic planes of β -phase CsPbI_3 perovskite structure, respectively, without observable peak shifts or additional peaks. To investigate the carrier dynamics of the

control and CBZ-modified films, steady-state photoluminescence (PL) and time-resolved PL (TRPL) spectroscopy characterizations were carried out. As shown in Fig. 3(b), the CBZ-modified film exhibits a significantly enhanced PL intensity compared to the control film, suggesting effective suppression of non-radiative recombination through CBZ surface treatment. This conclusion is corroborated by the TRPL curves in Fig. S8. By fitting the decay curves with a bi-exponential function, the average carrier lifetimes are prolonged from 5.94 to 11.36 ns for the CBZ-treated film. These results confirm that CBZ modification can effectively passivate defects in CsPbI_3 perovskite and mitigates trap-induced non-radiative recombination.

The ultraviolet-visible (UV-vis) spectra of the control and CBZ-modified CsPbI_3 films are recorded in Fig. 3(c), and it is difficult to discern significant differences in their absorption capabilities. The corresponding Tauc plots derived from these spectra are shown in Fig. S9, revealing that both films exhibit a nearly identical optical bandgap of ~ 1.68 eV, which is characteristic of the stabilized β - CsPbI_3 phase achieved with our chosen DMAI/CsI stoichiometric molar ratio of 1.0.^{43,44} Ultraviolet photoelectron spectroscopy (UPS) spectra were measured to analyze the surface energy levels of the CsPbI_3 films [Figs. 3(d) and 3(e)]. The work functions of the control and CBZ-modified perovskite films were determined to be 4.28 and 4.38 eV, respectively, with corresponding valence band (VB) positions at -5.35 and -5.57 eV. By subtracting the bandgap value, the conduction band (CB) positions were derived as -3.67 and -3.89 eV for the respective films [Fig. 3(f) and Table S1]. Apparently, the CBZ-modified film exhibits a more favorable energy level alignment, creating a favorable gradient that facilitates interfacial charge extraction while efficiently suppressing carrier recombination at the perovskite/PCBM interface.

Inverted devices were fabricated to evaluate the effect of the CBZ modification strategy on the photovoltaic performance of solar cells, as illustrated in Fig. 4(a). The typical current density-voltage (J - V)

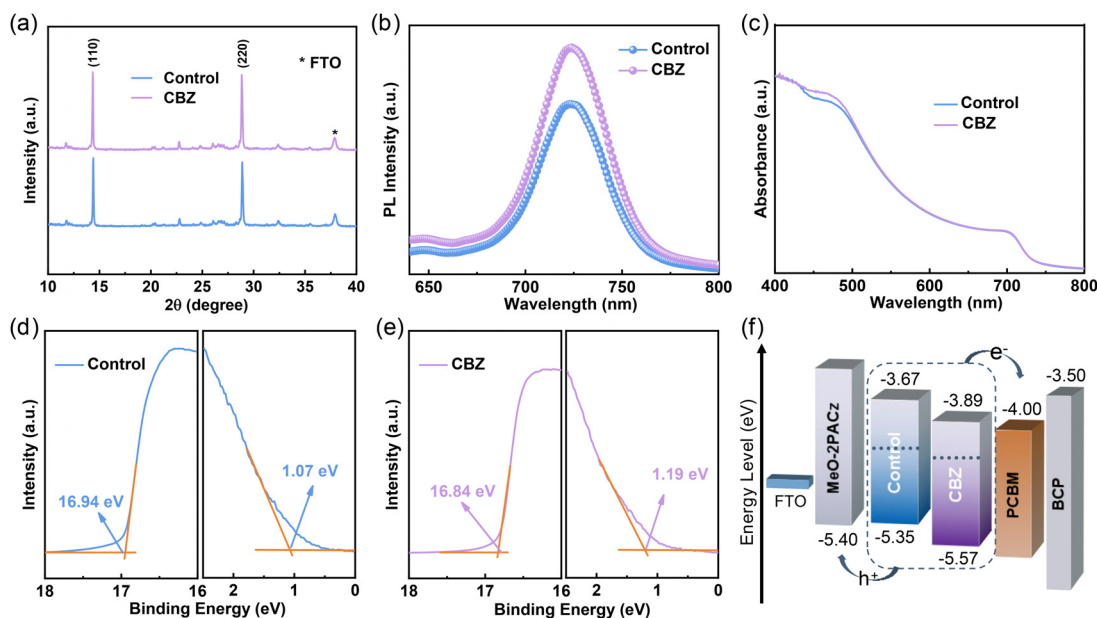


FIG. 3. (a) XRD patterns of control and CBZ-modified CsPbI_3 films. (b) Steady-state PL spectra. (c) UV-vis absorption spectra. (d) and (e) UPS spectra. (f) Energy level diagram of the functional layers.

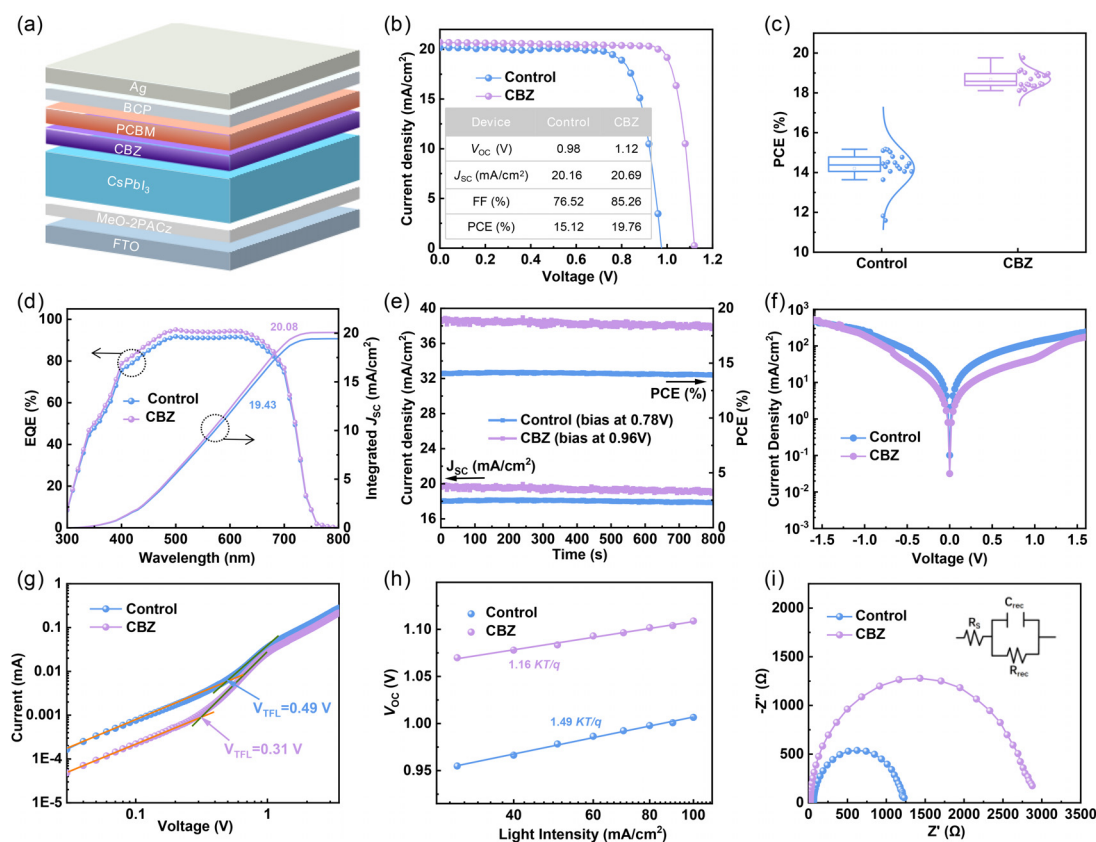


FIG. 4. (a) Device structure. (b) J - V curves of the champion PSCs. (c) Statistics of PCE values. (d) EQE spectra. (e) Stabilized PCE and J_{SC} of PSCs under MPP. (f) Dark J - V curves. (g) SCLC curves of hole-only devices. (h) Device V_{OC} dependence on light intensity of PSCs. (i) Nyquist plots.

characteristics of the devices were measured under standard AM 1.5G illumination (100 mW/cm^2). Figure S10 displays the J - V curves of the PSCs treated with various CBZ concentrations, and their statistical photovoltaic parameters are summarized in Fig. S11 and Table S2. The control device exhibits a moderate PCE of 15.12%, featuring a short-circuit current density (J_{SC}) of 20.16 mA/cm^2 , open-circuit voltage (V_{OC}) of 0.98 V, and FF of 76.52%. In contrast, when modified with a 0.2 mg/mL CBZ, the corresponding device obtains a champion PCE of up to 19.76%, with a J_{SC} of 20.69 mA/cm^2 , V_{OC} of 1.12 V, and FF of 85.26% [Fig. 4(b)]. Our device V_{OC} value is slightly low compared to the highest reported values. In future work, we could combine the CBZ interface strategy with advanced additive engineering techniques to further improve the V_{OC} value. It is worth mentioning that this outstanding FF represents one of the highest values ever reported in CsPbI₃ PSCs to date. We also fabricated n-i-p structured CsPbI₃ PSCs and confirmed that the beneficial effects of CBZ surface modification are applicable in the n-i-p devices (Fig. S12).

To evaluate the reproducibility of device performance, we summarized the PCEs of 20 individual devices without and with 0.2 mg/mL CBZ modification, and depicted a statistical box chart of the photovoltaic parameters in Fig. 4(c). The CBZ-modified PSCs show an enhanced reproducibility compared to the control devices. The external quantum efficiency (EQE) spectra were recorded to

validate the J_{SC} values obtained from J - V measurements [Fig. 4(d)]. The integrated J_{SC} values for the control and CBZ-modified PSCs are 19.43 and 20.08 mA/cm^2 , respectively, demonstrating excellent consistency with the J - V results. Furthermore, the steady-state power outputs of devices were tested through maximum power point (MPP) tracking for 800 s. As shown in Fig. 4(e), higher PCE and photocurrent are observed for the CBZ-modified device, verifying the benefit of CBZ modification. Additionally, CBZ modification effectively suppresses interfacial charge recombination, as evidenced by the dramatically reduced hysteresis observed in the modified devices (Fig. S13).

To elucidate the mechanisms underlying performance enhancement, especially considering the high value of FF, comprehensive characterizations were systematically carried out. The dark J - V characteristics [Fig. 4(f)] reveal that CBZ-modified devices exhibit significantly reduced leakage current in the dark state, signifying that charge carriers can be more effectively transported to corresponding electrodes due to the interface modification of CBZ. Space charge-limited current (SCLC) measurements were performed to quantify the defect density using hole-only devices.⁴⁵ As shown in Fig. 4(g), the trap-filled limit voltage (V_{TFL}) decreases from 0.49 to 0.31 V. It is clear that the CBZ-modified device exhibits a lower defect density value of $1.50 \times 10^{15} \text{ cm}^{-3}$ compared to $2.37 \times 10^{15} \text{ cm}^{-3}$ (control), indicating that CBZ modification can remarkably reduce the defect density and

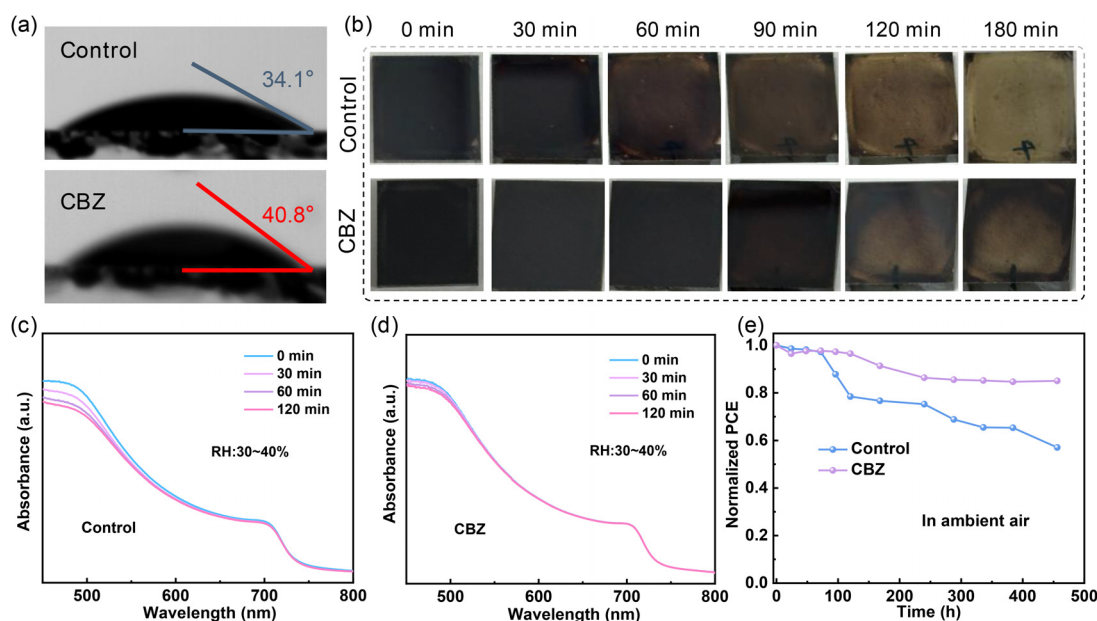


FIG. 5. (a) Contact angles of water on the control and CBZ-modified films. (b) The photograph evolution of CsPbI₃ films aged in air with an RH of ~50%. The UV-vis absorption spectra of (c) control, and (d) CBZ-modified films for different periods of time (30%–40% RH). (e) Long-term storage stability of the PSCs in air.

suppress trap-induced non-radiative recombination. The photocurrent decay profiles under short-circuit condition for PSCs are shown in Fig. S14, where a shorter decay time is observed for the CBZ-modified device, manifesting more efficient charge carrier transport processes. The carrier recombination dynamics of PSCs were further explored by conducting light intensity-dependence V_{OC} measurements, and the results are presented as a function of light intensity (I) on a logarithmic scale [Fig. 4(h)]. The ideality factor (n) of the devices was determined from the slope of V_{OC} vs light intensity using the following equation: $V_{OC} = nkT \ln(I)/q + C$, where k is the Boltzmann constant, T is the absolute temperature, q is the electron charge, C is a constant, and n reflects trap-assisted recombination.⁴⁶ The value of n for the CBZ-modified device decreased from 1.49 (control device) to 1.16, demonstrating effective suppression of trap-induced non-radiative recombination. Furthermore, the built-in potential (V_{bi}) extracted from the capacitance–voltage (C – V) curves is larger than that of the control films, providing stronger driving force for charge separation (Fig. S15).⁴⁷ Electrochemical impedance spectroscopy (EIS) data in Fig. 4(i) further corroborate these findings. The CBZ-modified device exhibits larger recombination resistance (R_{rec}), indicating suppressed charge recombination, and accelerated charge extraction.

To probe the synergistic effect on the stability of the CsPbI₃ film and device, we measured the contact angle of water droplets on the perovskite films [Fig. 5(a)]. The modified film exhibits a higher contact angle of 40.8° compared to 34.1° for the control film, indicating enhanced surface hydrophobicity that effectively protects the perovskite layer from moisture penetration. Furthermore, both control and CBZ-modified films were exposed to ambient air with ~50% relative humidity (RH) while monitoring their morphological evolution over time [Fig. 5(b)]. The control film exhibited extensive decomposition with macroscopic structural collapse after 180 min of exposure,

whereas the CBZ-modified film maintained remarkable morphological integrity under identical conditions, showing only minimal signs of degradation. The optical absorption of both films was continuously monitored under humid conditions [Figs. 5(c) and 5(d)]. The control sample presented remarkable absorption attenuation, while the CBZ-modified film displayed only minor absorption variations, indicating substantially improved film stability. The long-term device stability of both the unencapsulated control and CBZ-modified PSCs stored in air (30%–40% RH) was also examined [Fig. 5(e)]. The efficiencies of these devices showed negligible degradation within the first 100 h of storage. However, after 450 h, the control PSC underwent significant efficiency degradation, with its PCE dropping to 57% of the initial value, whereas the CBZ-modified device still preserved 85% of its initial PCE. These results conclusively establish CBZ modification as an effective strategy for simultaneously improving moisture resistance and long-term stability, primarily through defect passivation and enhanced hydrophobic protection.

In summary, we have developed a strategy for inverted CsPbI₃ PSCs that involves modifying the upper interface of the CsPbI₃ perovskite with CBZ to passivate defects. By leveraging its $-NH_2$ and $-C=O$ groups, CBZ can form strong coordination bonds with undercoordinated Pb^{2+} , passivating surface and GB defects. It also optimizes film morphology, interface energy level alignment as well as charge extraction while suppressing non-radiative recombination. Consequently, the CBZ-modified device achieves a champion PCE of 19.76% with an excellent FF of 85.26% and enhanced long-term stability in ambient air. This work validates the use of molecularly engineered modifiers as a feasible strategy for achieving efficient and stable inorganic PSCs.

See the [supplementary material](#) for the experimental section, Figs. S1–S15 and Tables S1 and S2.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Chenyu Zhou: Conceptualization (equal); Data curation (equal); Writing – original draft (equal). **Yuejin Zhu:** Funding acquisition (equal); Supervision (equal). **Xiaohui Liu:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal). **Jiayin Liu:** Data curation (equal); Investigation (equal). **Yiyang Qiu:** Data curation (supporting); Investigation (supporting). **Heng Qiao:** Data curation (equal); Investigation (equal). **Hubin Zeng:** Data curation (equal); Investigation (equal). **Jing Zhang:** Formal analysis (equal); Supervision (equal). **Like Huang:** Data curation (equal). **Ning Chen:** Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

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